FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2021

PAPER –2

TEST DATE: 18-01-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

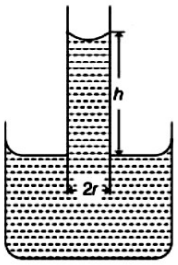
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A fine capillary tube of radius r dipped in a beaker containing water. The water rises in capillary upto height h. Let θ = angle of contact and ρ = density of water. Now, choose the correct options.



(A) Mass of liquid in capillary tube is $\pi r^2 h\rho$

(B) Mass of liquid in capillary tube is $\pi r^2\left(h+\frac{r}{3}\right)\rho$

(C) Surface tension of water is $T=\frac{hr\rho g}{2\cos\theta}$

(D) Surface tension of water in glass capillary is approximately $T=\frac{r\left(h+\frac{r}{3}\right)\rho g}{2}$

2. To study a relation of frequency and length of a given wire under constant tension using a sonometer, let you are performing an experiment in your school lab.
Now, choose the correct options.

(A) Tuning fork of known frequency is touched to wire for observing resonance

(B) Vibrations of wire are observed by paper riders

(C) Frequency of vibration of wire is $f=\frac{1}{LD}\sqrt{\left(\frac{T}{\pi\rho}\right)}$ (where, L = length of wire between bridges, D = diameter and ρ = density of wire)

(D) Frequency of tuning forks remains constant with temperature of lab

3. A cylindrical shaped wooden having mass m is floating inside a liquid with its axis is parallel to the vertical. The mass in depressed a little by a force and then force is removed, then

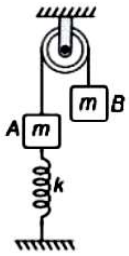
(A) mass will execute SHM with time period inversely proportional to radius of cylinder

(B) mass will execute SHM with time period $\frac{\pi r^2\sqrt{g}}{2m}$

(C) mass will not execute SHM

(D) mass will execute SHM with frequency $\frac{1}{2\pi}\sqrt{\frac{\pi r^2 g\rho}{m}}$

4. For the given figure, two blocks of equal mass m are connected by a light string that passes over a smooth pulley. The block A is connected by spring constant k. Initially spring is relaxed. Then, the block A is slowly pulled down a distance x and released. The magnitude of accelerations of the left and the right blocks immediately after releasing are a_1 and a_2, respectively.



(A) $a_1 = a_2$

(B) $a_1 = a_2$, if $k \le \frac{2mg}{x}$

(C) If $a_1=\frac{k}{m}x, a_2=0$

(D) $a_1 < a_2$, if $k > \frac{2mg}{x}$

5. A block is connected to a spring in an elevator. Elongation in spring is $x_1 = 4\sqrt{2}$ mm and $x_2 = 3\sqrt{2}$ mm, when it moves up and down with constant acceleration, respectively. Then,

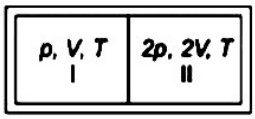
(A) acceleration of elevator is $\frac{g}{7}$ m/s^2

(B) acceleration of elevator is $\frac{g}{3}$ m/s^2

(C) if same elevator moves horizontally with same acceleration, then elongation will be 5 mm

(D) if same elevator moves horizontally with same acceleration, then elongation will be 10 mm

6. A partition divides a container having insulated walls into two compartments I and II. The same gas fills the two compartments, whose initial parameters are given. The partition is a conduction wall, which can move freely without friction. Which of the following statements is/are correct with reference to the final equilibrium position?



(A) The final pressure in the two compartments are equal

(B) Volume of compartment I is $\frac{3V}{5}$

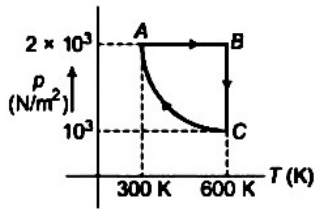
(C) Volume of compartment II is $\frac{12\,V}{5}$

(D) Final pressure in compartment I is $\frac{5p}{3}$

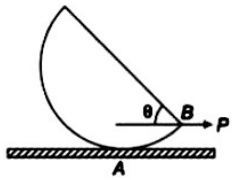
SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

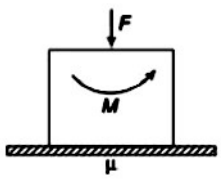
7. Two moles of a monoatomic ideal gas is taken through a cyclic process shown in p-T diagram. The process CA is represented as pT = constant. If efficiency of cycle is $1-\frac{3x}{12\ln 2+15}$, then find the value of x.



8. A short semi-circular right cylinder of radius r and weight w put on a horizontal surface and is pulled by a horizontal force P applied at the point B of the front edge. The flat face make angle θ with the horizontal plane just before sliding begins and coefficient of friction at the line of contact with surface is μ. Find the value of $7\sin\theta$, if μ is $1/\pi$.

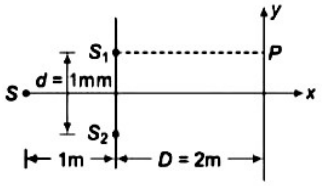


9. A shaft of diameter D is resting on a support as shown in the figure. The shaft exerts a total axial force F. Assuming the coefficient of friction is μ, the twisting moment M necessary to cause rotary motion to impend is μ FD/N. Find the value of N.

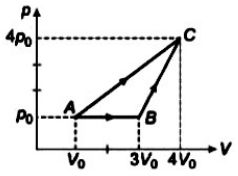


10. A photoelectric plate is initially exposed to a spectrum of hydrogen gas excited to second energy level. Later when the same photoelectric plate is exposed to radiation from some unknown hydrogen like gas, excited to second energy level, it is found that the de-Broglie wavelength of the photoelectrons, now ejected has decreased $\sqrt{6.1}$ times. For this new gas difference of energies of first Lyman series and Balmer series, limit is found to be two times, the ionisation potential of the hydrogen atom. Find the value of Z for unknown gas.

11. In a modified YDSE, the source S of wavelength 5000Å oscillates about axis of set up according to the equation, $y = 0.5\sin\left(\frac{\pi}{6}\right)t$, where y is in millimetre and t in second. At what time t will the intensity at P, a point exactly in front of slit S_1, be maximum for the first time?



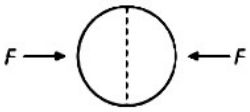
12. A certain quantity of ideal gas takes up 56 J of heat in the process AB and 360 J in the process AC. What is the number of degree of freedom of the gas?



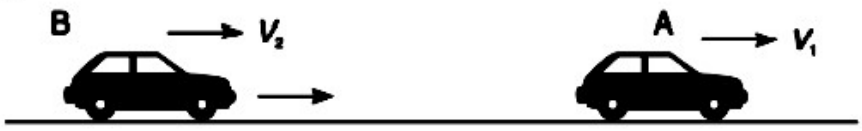
SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

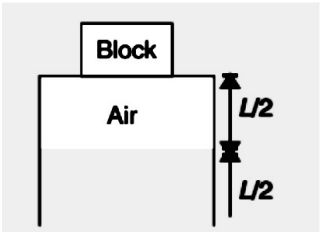
13. A uniformly charged thin spherical shell of radius R carries uniform surface charge density 6 per unit area. It is made of two hemispherical shells, held together by pressing them with force F. The value of F will be (Take, $\pi\sigma^2R^2 = 4\varepsilon_0$)



14. A police car B is chasing a culprit's car A. Car A and B are moving at constant speed V1 = 108 km/hr and V2 = 90 km/hr respectively along a straight line. The police decide to open fire and a policeman starts firing with his machine gun directly aiming at car A. The bullet has a velocity u = 305 m/s relative to the gun. The policeman keep firing for an interval of T_0 = 20 s. The Culprit experiences that the time gap between the first and the last hitting his car is Δt is

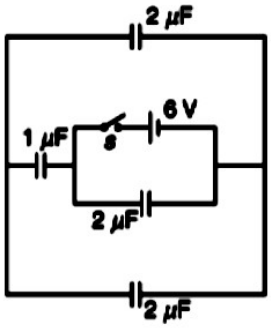


15. A light cylindrical tube of length L = 1.5 m and radius $r = \frac{1}{\sqrt{\pi}}m$ is open at one end. The tube containing air is inverted and pushed inside water as shown in figure. A block made of material of relative density 2 has been placed on the flat upper surface of the tube and the whole system is in equilibrium. Neglect the weight of air inside the tube the volume of block placed on the tube is



16. A transverse harmonic wave of amplitude 4 mm and wavelength 1.5 m is travelling in positive x direction and at x = 1.0 m is at y = + 2 mm and is travelling in positive y direction, then x-coordinate of the nearest particle (x > 1.0 m) which is at its positive extreme at this instant is

17. Find charge supplied by the cell after the switch is closed (in μC)



18. Sunray pass through a pinhole in the roof of a hut and produce an elliptical spot the floor. The minor and major axes of the spot are 6 cm and 12 cm respectively. The angle subtended by the diameter of the sun at our eye is 0.5°. Height of the roof in meters is

Chemistry

PART – II

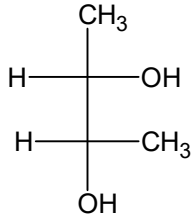
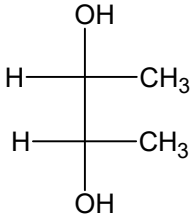
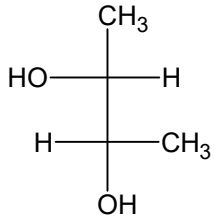
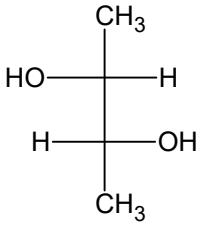
SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

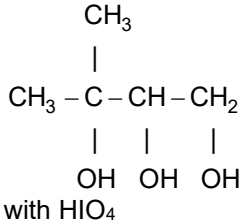
19. Select correct statements regarding σ and π-bonds:
(A) σ-bond lies on the line joining the nuclei of bonded atoms
(B) π-electron cloud lies on either side to the line joining the nuclei of bonded atoms
(C) ($2p\pi$-$3d\pi$) Pi-bond is stronger than ($2p\pi$-$3p\pi$) Pi-bond
(D) σ-bond has primary effect to decide direction of covalent bond, which π-bond has no primary effect in direction of bond.

20. Complex compound $[Co[SCN)_2(NH_3)_4]Cl$ exhibits:
(A) Ionization isomerism
(B) Geometrical isomerism
(C) Optical isomerism
(D) Linkage isomerism

21. Which of the following metal(s) is/are commercially extracted by self reduction method from their corresponding ore?
(A) Cu
(B) Fe
(C) Pb
(D) Hg

22. Which of the following structures is/are meso-2,3-butanediol
(A)

(B)

(C)

(D)


23. Which of the following compounds are formed by the oxidation of



with HIO_4

(A) $CH_3 - CHO$

(B) CH_3COCH_3

(C) $H - COOH$

(D) $HCHO$

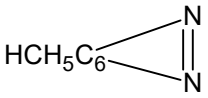
24. $C_6H_5 - CH_2 - I \xrightarrow[\Delta]{NaN_3} \text{Products}$

Reaction is assumed to involve nitrene as intermediate, then various possible product are:

(A) $C_6H_5CH_2NH_2$

(B) $C_6H_5N = CH_2$

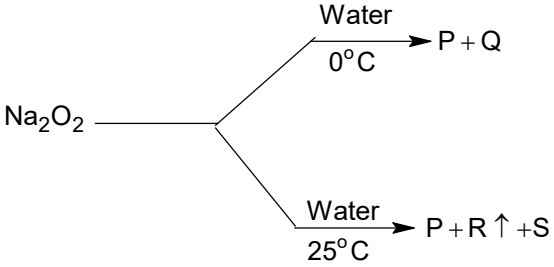
(C) $C_6H_5CH = NH_2$

(D) 

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

25. Calculate sum of bond order between same bonded atoms in Q and R compounds.



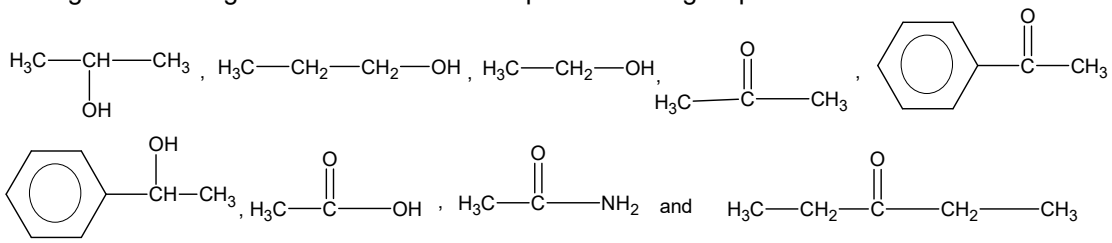
26. Calculate value of $|x + y - z|$ for the following silicate $[Si_xO_{y+z}]^{z-}$ anion.



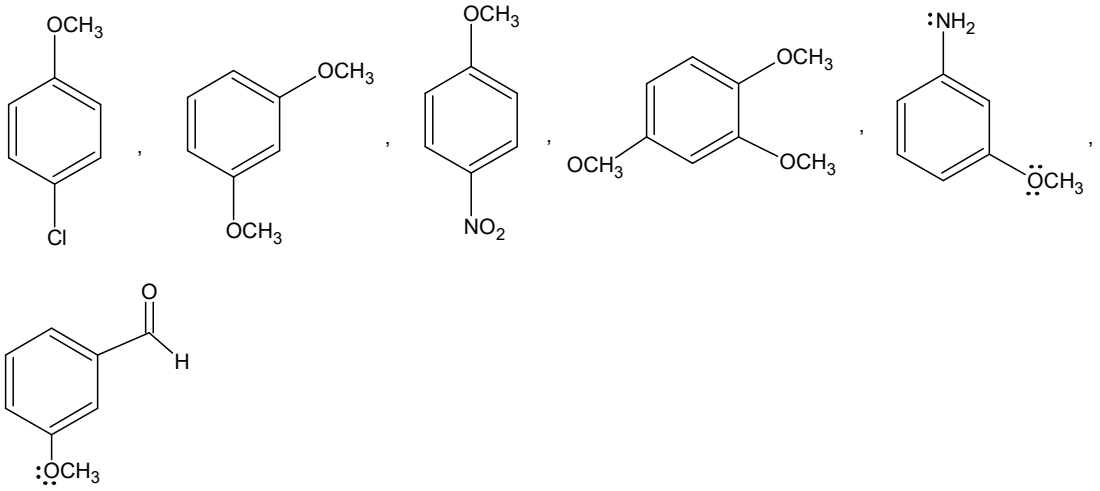
27. How many stereoisomers does the following complex have?
$[FeF_2Cl_2Br_2]^{3+}$

28. A hydrocarbon $C_xH_{12}(g)$ and $O_2(g)$ were taken in a closed flask at 400K. When sparked they reacted to form $CO_2(g)$ and $H_2O(g)$ and no reactants left after the reaction. The pressure in the flask is increased by 20%. The value of 'x' is

29. Among the following the total number of compound which give positive iodoform test is:



30. Examine the structural formulas shown below and identify how many compounds will show coupling reaction with diazonium salts faster than anisole $(Ph-\ddot{O}-CH_3)$



SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

31. A transition metal X forms an oxide of formula X_2O_3. It is found that only 50% of X atoms in this compound are in the +3+ oxidation state. The only other stable oxidation states of X are +2 and +5. What percentage of X atoms is in the +2 oxidation state in this compound?

32. One mole of an non-ideal gas undergoes a change of state from (2.0atm, 3.0 L, 95K) to (4.0 atm, 5.0L, 245 K) with a change in internal energy, $\Delta U = 30.0\,L$ atm. The enthalpy change (ΔH) of the process in L-atm is

33. A container of capacity V L contains an equilibrium mixture that consists of 2 moles each of PCl_5,PCl_3 and Cl_2 (all as gases) The pressure is 30.3975 kPa and temperature is TK. A certain amount ('x' mole) of Cl_2() is now introduced keeping the pressure and temperature constant, until the equilibrium volume becomes 2V L. the value of '90x' is

34. The specific conductivity of a saturated solution of AgCl is 2.80× 10^{-4} mho m^{-1} at 25°C. If $\lambda^{o}_{Ag^+} = 6.19\times10^{-3}$ mho m^2 mol^{-1} and $\lambda^{o}_{Cl^-} = 7.81\,mho m^2\ mol^{-1}$, the solubility of silver chloride (in order of $10^{-5}\,gl^{-1}$) at 25°C, is

35. Vapoure pressure of solution containing 6g of a non-volatile solute in 180g water is 20 torr. If 1 mole of water is further added, vapour pressure increases by 0.02 torr. The molar mass (in g/mol) of the non-volatile solute is

36. For a gaseous reaction at 300K, $\Delta H - \Delta U = -49.8kJ$, assuming that R = 8.3 $JK^{-1}\,mol^{-1}$, Δng is

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Suppose $I_1 = \int_0^{\pi/2} \cos\left(\pi \sin^2 x\right) dx$, $I_2 = \int_0^{\pi/2} \cos\left(2\pi \sin^2 x\right) dx$, $I_3 = \int_0^{\pi/2} \cos\left(\pi \sin x\right) dx$ then

(A) $I_1 = 0$

(B) $I_1 + I_3 = 0$

(C) $I_1 + I_2 + I_3 = 0$

(D) $I_2 = I_3$

38. The equations of line(s) through (2, 3) from which the circle $x^2 + y^2 = 25$ intercept a chord of length 8 unit is/are?

(A) $x = 2$

(B) $y = 3$

(C) $9x + 11y - 51 = 0$

(D) $12x + 5y - 39 = 0$

39. If $y = \log_7\left(\log_7 x\right)$, then $\frac{dy}{dx}$ equal

(A) $\frac{\log_7 e}{x \log_e x}$

(B) $\frac{1}{x \log_e x \log_e 7}$

(C) $\frac{1}{\log_e (7x)^x}$

(D) $\frac{\log_7 e . \log_x e}{x}$

40. A curve y = f(x) has the property that the perpendicular distance of the origin from the normal at any point P of cone is equal to the distance of point P from x-axis. Then the differential equation of curve is

(A) Homogenous

(B) Can be converted into linear differential equation with some suitable substitutes

(C) Is the family of circles touching the x-axis at origin

(D) Family of circles touching y-axis at origin

41. If $\int x \log\left(1 + x^2\right) dx = \phi(x).\ln\left(1 + x^2\right) + \psi(x) + c$ then

(A) $\phi(x) = \frac{1 + x^2}{2}$

(B) $\psi(x) = \frac{x^2 + 1}{2}$

(C) $\psi(x) = -\dfrac{x^2+1}{2}$

(D) $\psi(x) = -\dfrac{x^2+1}{3}$

42. Equation(s) of tangent(s) to hyperbole $\dfrac{x^2}{4} - \dfrac{y^2}{3} = 1$ at point(s), where it intersect the parabola, $y^2 = \dfrac{9}{4}x$ are

(A) x – y + 1 = 0
(B) x – y – 1 = 0
(C) x + y – 1 = 0
(D) x + y + 1 = 0

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. If the value of $\sum_{r=1}^{5}\left(x^r + \dfrac{1}{x^r}\right)^2$, where $x^2 + x + 1 = 0$

44. Number of points, where f(x) = maximum $f(x)\left\{\text{sgn}(x), -\sqrt{9-x^2}, x^3\right\}$ is continuous but not differentiable

45. The distance of point (1, –2, 3) from plane x – y + z= 5 measured parallel to line $\dfrac{x}{2} = \dfrac{y}{3} = \dfrac{z-1}{-6}$ is

46. If f(x) = $\left[9^x - 3^x + 1\right] \forall x \in (-\infty, 1)$ ([.] denotes the greatest integer function) then number of elements in range of f(X) are

47. An artillery target may be either at point I with probability $\dfrac{8}{9}$ or at point II with probability $\dfrac{1}{9}$. we have 21 shells each of which can be fired either at point I or II. Each shell may hit the target independently of the other shell with probability $\dfrac{1}{2}$. If α shells must be fired at point I to hit the target with maximum probability, than $\dfrac{\alpha}{2}$ equals

48. The area bounded by curve y = x + sin x and its inverse function between the ordinates x = 0 and x = 2π is 4α find α ?

SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

49. If AM, GM, HM of first and last term of series 100, 101, 102, …. n – 2, n –1, n are the terms of series itself (100 < n $\leq$ 500). Find n ? (AM, GM, HM indicates Arithmetic mean, Geometric Mean, Harmonic Mean respectively)

50. How many 4 letter words can be made using letter of word TATTERS ?

51. Find imaginary port of $\begin{vmatrix} -5 & 3+5i & \frac{3}{2}-4i \\ 3-5i & 8 & 4+5i \\ \frac{3}{2}+4i & 4-5i & 9 \end{vmatrix}$

52. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ ard $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 2$

Then $\left[\vec{a}\vec{b}\vec{c}\right]$ cos 45º equals

53. The curve $y = ax^3 + bx^2 + cx + 5$ touches x-axis at point P(–2, 0_) and cuts y-axis at a point Q where its gradient is 3 find |c – a – b| ?

54. If $\sin 3\theta + \cos 2\theta = 0$ for $\theta = \left[0, \frac{3\pi}{2}\right]$

Number of value of θ can be